

11.3 A 40-nm, 2M-Cell, 8b-Precision, Hybrid SLC-MLC PCM Computing-in-Memory Macro with 20.5 - 65.0TOPS/W for Tiny-AI Edge Devices

Win-San Khwa^{*1}, Yen-Cheng Chiu^{*2}, Chuan-Jia Jhang², Sheng-Po Huang², Chun-Ying Lee², Tai-Hao Wen², Fu-Chun Chang², Shao-Ming Yu¹, Tung-Yin Lee¹, Meng-Fan Chang^{1,2}

¹TSMC Corporate Research, Hsinchu, Taiwan

²National Tsing Hua University, Hsinchu, Taiwan

*Equally Credited Authors (ECAs)

Efficient edge computing, with sufficiently large on-chip memory capacity, is essential in the internet-of-everything era. Nonvolatile computing-in-memory (nvCIM) reduces the data transfer overhead by bringing computation closer, in proximity, to the memory [1-4]. While the multi-level cell (MLC) has higher storage density than the single-level cell (SLC). A few MLC or analog nvCIM designs had been proposed, but they either target simpler neural-net models [5] or are implemented using a less area-efficient differential cell [6]. Furthermore, representing the entire weight vector using one storage type does not exploit the drastic accuracy difference between the upper and the lower bits.

Figure 11.3.1 illustrates the three major challenges for a hybrid SLC-MLC nvCIM macro: (1) the tradeoff between signal margin and memory density; (2) WL activation-dependent signal margin degradation; and (3) the low zero-skipping probability resulting from processing the input in-order. The following three schemes are proposed: (1) a hybrid SLC-MLC for 8b weight storage to strike a balance between accuracy, energy efficiency (EF), and weight density; (2) a voltage-swing remapping voltage sense amplifier (VSR-VSA) to enhance the signal margin; and (3) an input-reordering (IN-R) scheme to improve throughput and EF. The proposed 2M-cell phase-change-memory (PCM) based nvCIM macro is fabricated using a 40nm technology. The macro supports 1/4/8b-input and 2/4/8b-weight bitwize multiply-accumulate (MAC) computation. This work presents the first hybrid SLC-MLC nvCIM macro that leverages SLC's signal margin and MLC's area density. This macro achieves a 3.25 - 15.9ns access time, and an EF of 20.5 and 65.0TOPS/W for an 8b IN - 8b W - 19b OUT using 2× SLC + 3× MLC and 4× MLC setups. Using a FoM, which incorporates weight density and sensing margin ($FoM = IN-W-EF \cdot D_w / Avg. MAC \text{ levels}$), shows this work to be 4× better than prior work; where, D_w represents the weight storage area efficiency (number of weight bits / number of devices used) and the average MAC level is the number of MAC levels used in a design's computational dynamic range, representing a normalized signal margin.

Figure 11.3.2 shows the proposed hybrid SLC-MLC PCM nvCIM architecture. To perform a MAC operation, 1b inputs are applied on eight WLs simultaneously; bit-serial assertion over 8 cycles is used to extend this to an 8b input on each WL. The 8b weight is represented in 2's complement, using five neighboring PCM cells on the same WL. The upper two weight bits are stored in SLC format to ensure readout accuracy, while the lower six bits are stored in MLC format for area efficiency. Each column has a current-controlled voltage-metric block to generate the V_{MAC_IN} with a fixed current (I_m) via a bias NMOS (N_{BIAS}). V_{MAC_IN} is the voltage resulting from the eight activated PCM cells in the same column that is divided into 8 and 24 MAC values (MACV) for SLC (MACV=0 to 8) and MLC (MACV=0 to 24) columns, respectively. This voltage is sensed by the VSR-VSA to generate a 4b digital output. Lastly, to incorporate the place values of the 8b weight, the 4b VSR-VSA outputs across five columns are combined by the digital shifter and adder to obtain the final 19b digital outputs of an 8b-input times 8b-weight with 8 accumulations.

Figure 11.3.3 shows the operation and the block diagram of the proposed IN-R scheme. In a conventional approach, the 256 inputs are divided into 32 groups and arranged in the order of how their corresponding weights are stored on the 256 PCM rows. Since the input values are randomly distributed, most input groups would still require 8 phases to apply their 8b input onto the WL bit-serially, as shown in Fig. 11.3.1. In the proposed scheme, the inputs are reordered to increase the probability of group sparsity. As a result, the number of input phases is reduced: shortening the total operation time and reducing energy used. The 256-input groups are categorized into three types – high input value ($H-IN \geq 64$, represented by 7 or 8b), low input value ($L-IN < 64$, represented by 6 or less bits), and zero input value ($Z-IN=0$). The Z-IN type is skipped to save input phases. During input fetching, the H-IN and L-IN types are grouped using the 6b and 8b registers; when eight entries are collected, the inputs are applied on the WL. Because the L-IN groups require no more than six phases, two of the upper bit phases can be skipped.

Figure 11.3.4 shows the VSR-VSA operation: comprising of a 2b Flash ADC and a voltage remapping circuit. In standby mode, the WL is on and V_{MAC_IN} (V_{IN}) starts to develop according to the MAC result from the eight PCM cells in a column. SW1 is turned on to deliver V_{MAC_IN} to node IN, and SW2 and SW3 are off to isolate the V_{shift} selector. In phase-1 (PH1), SEN is on to detect V_{IN} and generate OUT[3:2], while V_{MAC_IN} (V_{IN}) continues to develop. In phase-2 (PH2), the V_{shift} selector selects the shift voltage ($V_{shift[n]}$) based on OUT[3:2], and SW3 is turned on to drive the AS node (V_{AS}) to the chosen $V_{shift[n]}$ voltage. In phase-3 (PH3), V_{MAC_IN} development stabilizes, SW1 and SW3 are off, and SW2 is on to make V_{AS} transit from $V_{shift[n]}$ to V_{MAC_IN} . The voltage difference ($V_{MAC_IN} - V_{shift[n]}$) is coupled to the IN node through C_{AS} ; making $V_{IN} = 2V_{MAC_IN} - V_{shift[n]}$ and resulting in a 2× margin enhancement. In phase-4 (PH4), SEN is turned on again to generate OUT[1:0]. Finally, the digital shifter and adder accumulates OUT[3:0] across the five IOs to output a 19b MACV (8b IN - 8b W - 8 accumulations).

Figure 11.3.5 shows the simulated performance of the proposed schemes. The IN-R scheme improves the energy efficiency (EF) by 1.58× under a 60% input activity sparsity. With VSR-VSA, the average signal margin (ASM) of a SLC-MLC column is enhanced by 1.8×. These simulations include non-linear PCM IV characteristics, its resistance variation and pattern-dependent variation. We calculate the effective ASM (ASM_{EFF}) by taking the sum of products between ASM and its corresponding place value ($ASM_{EFF} = \sum 2^n ASM[n]$), multiply it by EF, and divide the result by the number of devices used to represent the 8b weight (e.g. $FoM = ASM_{EFF} \times EF / 5$ for 2 SLCs + 3 MLCs). While a pure-SLC achieves the best ASM and a pure-MLC achieves the best area density and EF, the proposed hybrid 2×SLC + 3×MLC configuration is the best trade-off point.

Figure 11.3.6 presents the measurement results of the proposed macro. The measured CIFAR-100 inference accuracy shows an 0.81% degradation for 2×SLC + 3×MLC, and a 6.85% degradation for 4×MLC. The measured 8b IN - 8b W - 19b OUT EF are 20.5 and 65.0TOPS/W for a 2×SLC + 3×MLC with a 55% input sparsity and a 4×MLC with a 90% input sparsity, respectively. The accuracy-EF-area tradeoff suggests that a hybrid 2×SLC + 3×MLC achieves the best balance, while a 4×MLC should be considered for the best EF at the cost of accuracy (e.g. low-power mode). The Shmoo plot demonstrates a 14.3ns access time for an 8b IN - 8b W - 19b OUT computation at 0.95V, while the logic analyzer waveform shows correct operation. This work achieves a 4× better FoM than earlier nvCIM work. Figure 11.3.7 shows the die photo and a summary table.

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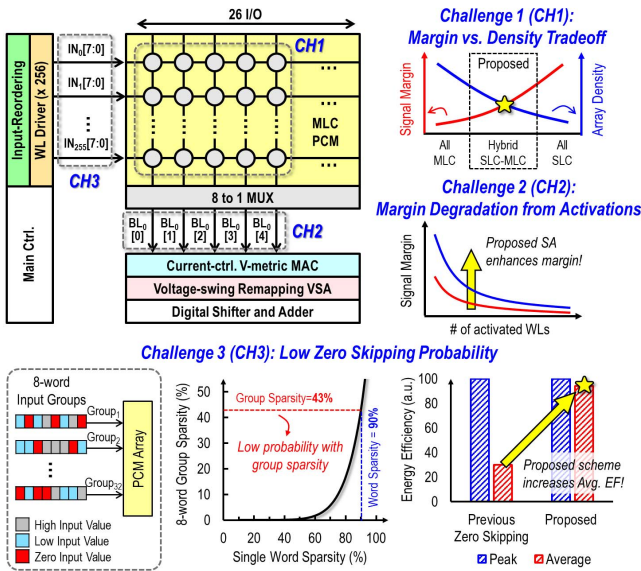


Figure 11.3.1: Hybrid SLC-MLC nvCIM macro challenges – (1) margin vs density tradeoff, (2) margin vs activation tradeoff, and (3) low zero skipping probability.

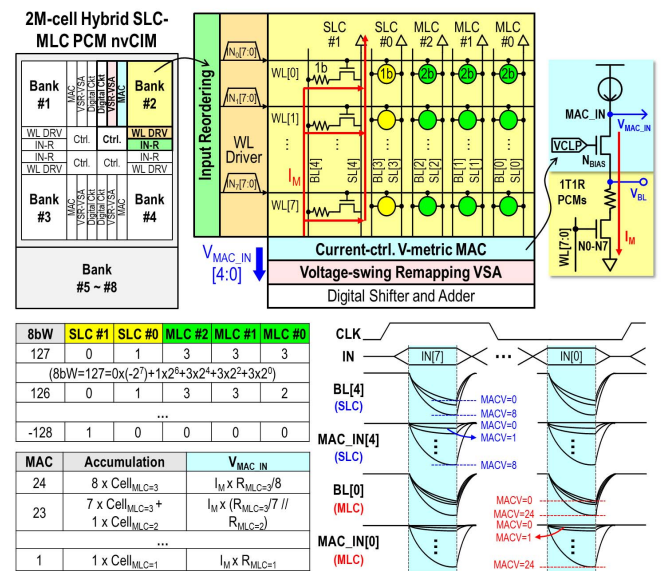


Figure 11.3.2: Hybrid SLC-MLC PCM nvCIM architecture features bit-serial WL assertions with digital shifter and adder to perform a MAC operation.

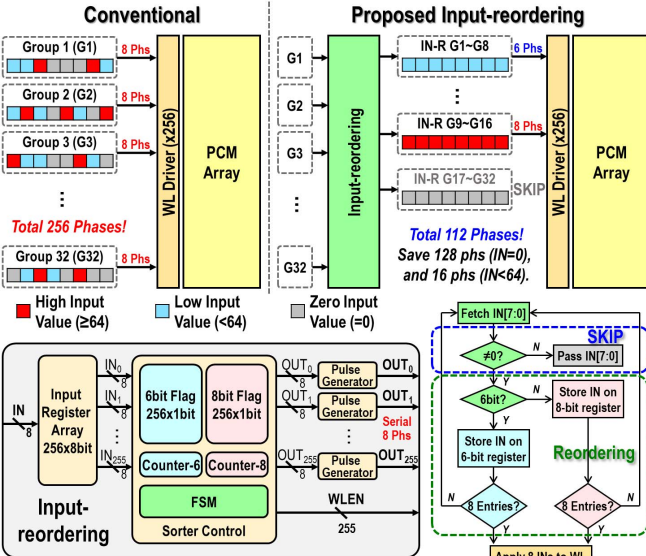


Figure 11.3.3: The input-reordering (IN-R) scheme reorders the input vectors according to their input values (high, low, or zero).

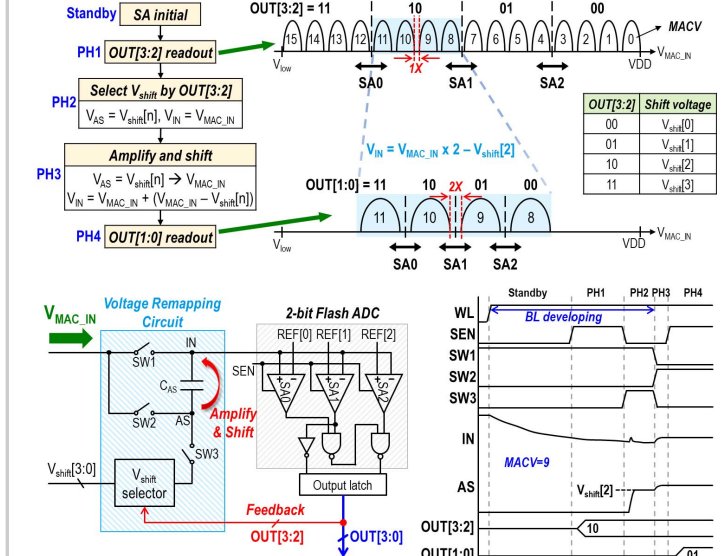


Figure 11.3.4: The voltage-swing remapping VSA (VSR-VSA) scheme amplifies the voltage difference through capacitive coupling of C_{AS} .

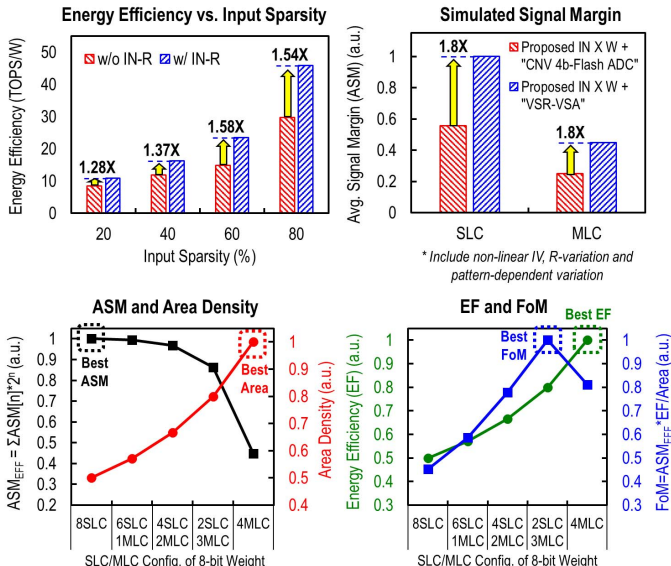


Figure 11.3.5: Simulated performance shows the improvements of the proposed schemes and compares between different SLC/MLC configurations.

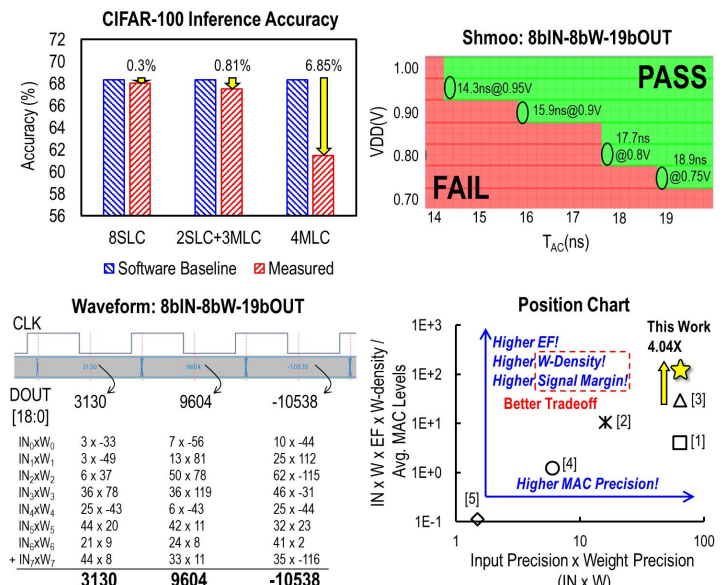


Figure 11.3.6: Measurement results and comparison.



Chip Summary		
Technology	40nm CMOS Logic Process	
PCM	Foundry 1T1R Hybrid PCM	
Testchip Size	18mm ²	
Capacity	2M Cells (8 Sub-bank)	
Sub-bank	256 rows x 1024 columns	
Performance		
Access Time (ns) @ 0.9V	1bIN-2bW-5bOUT	3.25
	4bIN-4bW-11bOUT	8.6
	8bIN-8bW-19bOUT	15.9
Throughput (TOPS)	1bIN-2bW-5bOUT	[†] 3.9
	4bIN-4bW-11bOUT	[†] 0.829
	8bIN-8bW-19bOUT	[†] 0.475
Energy Efficiency (TOPS/W)	1bIN-2bW-5bOUT	[†] 261 - [‡] 718
	4bIN-4bW-11bOUT	[†] 57.6 - [‡] 182
	8bIN-8bW-19bOUT	[†] 20.5 - [‡] 65
Inference Accuracy (%)	8bIN-8bW-19bOUT	[‡] 3 CIFAR-10: 91.89
		[‡] 3 CIFAR-100: 67.53

^{*1} Performance @ 0.85V in hybrid 2SLC-3MLC with IN-R and 55% sparsity.

^{*2} Performance @ 0.75V in 4MLC with IN-R and 90% sparsity.

^{*3} Performance @ 0.9V in hybrid 2SLC-3MLC with ResNet-20 model.

Figure 11.3.7: Die micrograph and summary table.